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M#: M21185543

1. What did you learn?

In this lab I learned about how to apply the concept of threads to solve a mathematical problem, as well as the concept of applying multiple threads to break up work into "chunks", which is a way that humans sometimes deal with large tasks in the real world. I also learned about the potential importance of tracking programs in real time because it may help with improving the efficiency of the program.

2. What challenges did you encounter, if any?

I did not experience any challenges with this activity as I was able to reference the solution when I got stuck after attempting the problem myself and the author explained their work in great detail, in my opinion.

Test Run of Program with 5 threads

ThreadDriver.java MyThreads.java

```
        for (int i = 0; i < threadCount; i++) {
            threads[i].join(); // will throw an exception so the other threads can work together
        }

        // preparing results
        int maxValue = 0; // these will be replaced with our answers
        int maxDivisors = 0;

        for (int i = 0; i < helpers.length; i++) { // traverse through the array
            MyThreads helperThread = helpers[i];
            if (helperThread.getMaxDivisors() > maxDivisors) {
                maxDivisors = helperThread.getMaxDivisors();
                // each helper has a different result depending on the range, if statement helps us find the best helper
                maxValue = helperThread.getFinalNumber();
            }
        }
        long endTime = System.currentTimeMillis(); // for final output.
        System.out.println("Number with most divisors: " + maxValue);
        System.out.println("Number of divisors: " + maxDivisors);
```

Console

```
<terminated> ThreadDriver [Java Application] /Users/javaughnstevens79/Downloads/jdk-25.0.2.jdk/Contents/Home/bin/java  (Apr 24, 2026, 7:06:52 PM – 7:06:54 PM elapsed 0:00:01
How many threads do you want to use?
5
Number with most divisors: 83160
Number of divisors: 128
Elapsed time: 87 ms or 0.087 s
```